Relational Model

Relational Data Model

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Multiple relations.
No pointers.
Connections via repeated data.

Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

Product

p_name	p_price	p_man	p_address	p_city	p_state
Gizmo	$19.99	GizmoWorks	123 Gizmo St.	Houston	TX
Powergizmo	$39.99	GizmoWorks	123 Gizmo St.	Huston	TX
Widget	$19.99	WidgetsRUs	20 Main St.	New York	NY
HyperWidget	$203.99	Hyper	1 Mission Dr.	San Francisco	CA
VaporWare	NULL	NULL	NULL	NULL	NULL

Just one?

Product

p_name	p_price	p_man
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Manufacturer

m_name	m_address	m_city	m_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

Relation vs. Table

Relation

- Part of the model
- No duplicates (set)
- Unordered

Table

- Part of the implementation
- Allows duplicates (multiset)
- Allows ordering

Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

Attribute / Column / Field

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
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VaporWare	NULL	NULL

A typed data entry for each row.

heading = attribute name

#attrs. = arity = degree

Tuple / Row / Record

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A single entry.

Each tuple is unordered: a mapping from attribute name to value.

#tuples = size

Cell / Field

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VaporWare	NULL	NULL

Schema

Product

prod_name	prod_price	prod_manufacturer
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Powergizmo	$39.99	GizmoWorks
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VaporWare	NULL	NULL

Product (prod_name, prod_price, prod_manufacturer)

Product (prod_name: **string**, prod_price: **money**, prod_maufactuer: **string**)

Instance / Realization

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

schema + a particular set of data

Mathematical Definition

- *Relation* $R(a, b, \ldots, z) \subseteq A \times B \times \cdots \times Z$

  where $a \in A, b \in B, \ldots, z \in Z$

Set of tuples for which the relation is *true*.